SALEH RAMADAN

* Digital Communications *

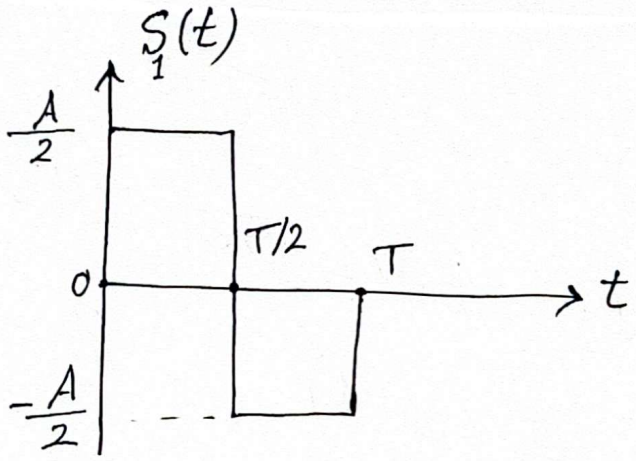
"Computer"

"Sheet 3 solution"

***Problem 1:**

*** Case 1:**

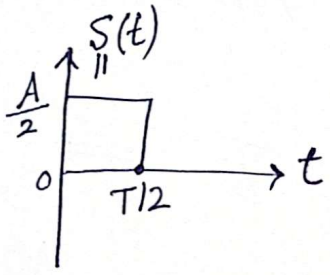
a)



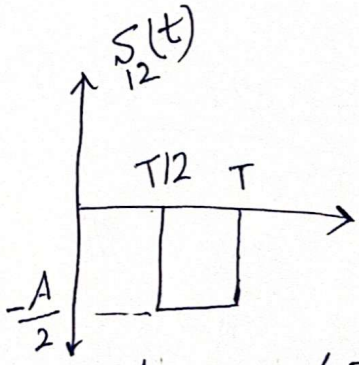
$$\cdot\ S_1(t) = \begin{cases} \frac{A}{2} & 0 < t < \frac{T}{2} \\ \frac{-A}{2} & \frac{T}{2} < t < T \end{cases}$$

$$h_1(t) = S_1(T-t)$$

$S_1(t) \Longrightarrow$



$\oplus$



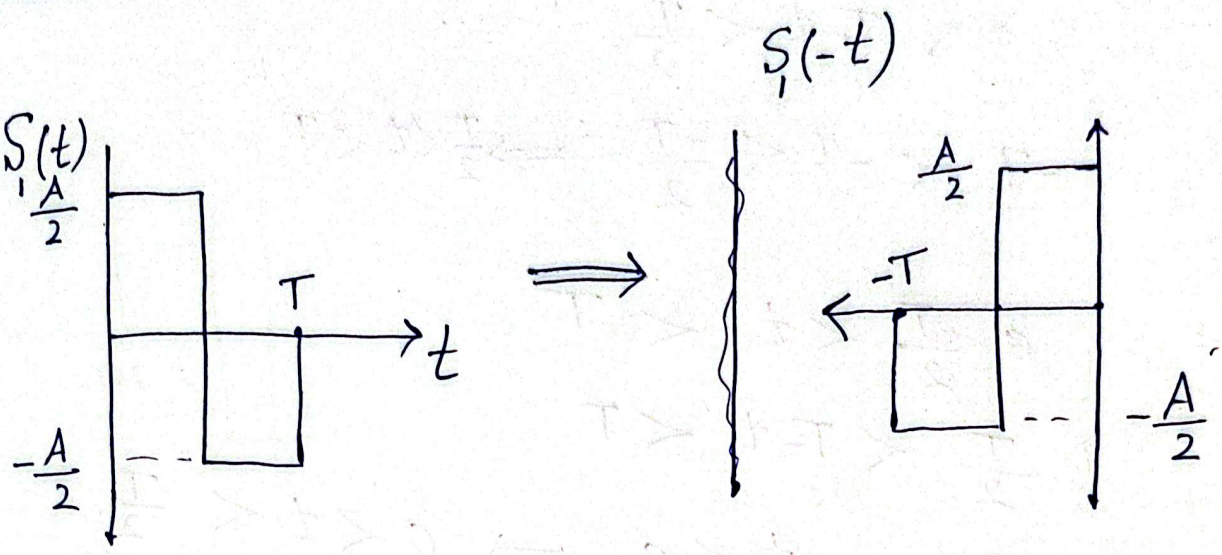
$$S_{11}(t) = \frac{A}{2}\,\text{Rect}\left[\frac{t - T/4}{T/2}\right], \quad S_{12}(t) = \frac{-A}{2}\,\text{Rect}\left(\frac{t - \frac{3T}{4}}{T/2}\right)$$

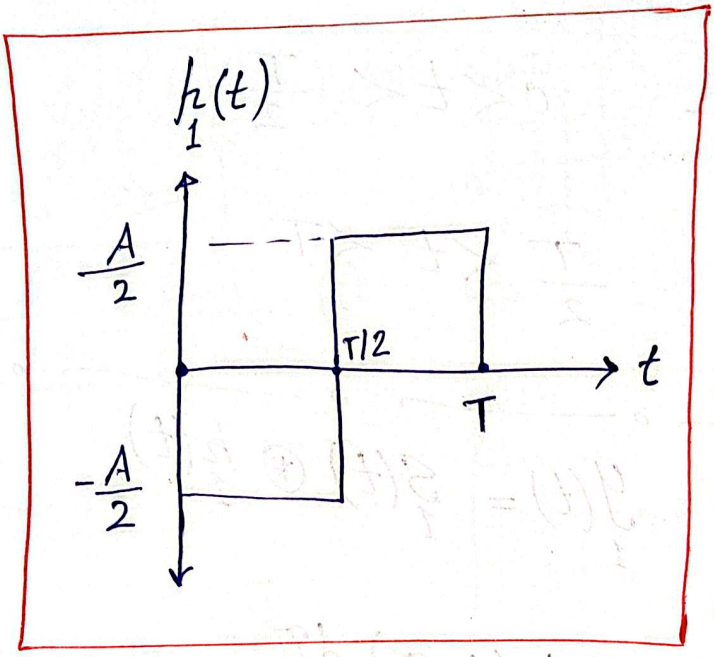
$$S_1(t) = \frac{A}{2}\left[\text{rect}\left(\frac{t - T/4}{T/2}\right) - \text{rect}\left(\frac{t - 3T/4}{T/2}\right)\right]$$

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$$h_1(t) = \frac{A}{2}\left[\text{rect}\left[\frac{\frac{3}{4}T - t}{T/2}\right] - \text{rect}\left[\frac{T/4 - t}{T/2}\right]\right]$$

→ Sketch of $h_1(t)$:





كنت ممكن توصل لـ Sketch من هنا

$$S_1(t) = \begin{cases} \frac{A}{2} & 0 < t < \frac{T}{2} \\ \frac{-A}{2} & \frac{T}{2} < t < T \end{cases}$$

شيل كل t وحط T-t ، فالـ Regions هتتغير ال

$\frac{A}{2}$:

$$0 < t < \frac{T}{2}$$

$$0 < T-t < \frac{T}{2}$$

$$-T < -t < \frac{-T}{2} \Rightarrow \frac{T}{2} < t < T$$

$\frac{-A}{2}$:

$$\frac{T}{2} < t < T$$

$$\frac{T}{2} < T-t < T$$

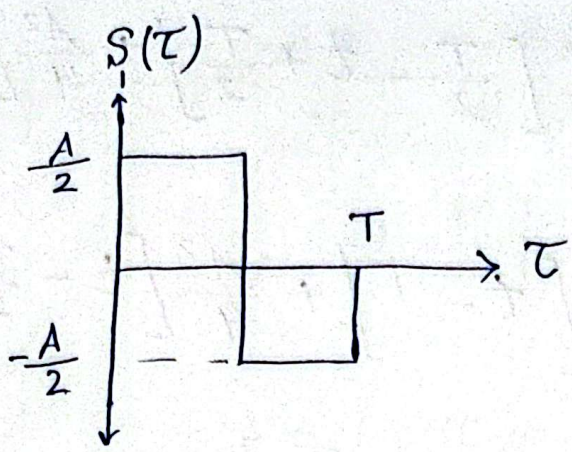
$$\frac{-T}{2} < -t < 0 \Rightarrow 0 < t < \frac{T}{2}$$

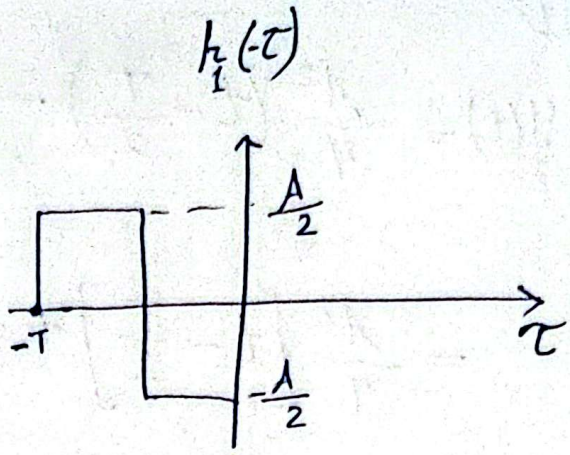
$$h_1(t) = \begin{cases} \frac{-A}{2} & 0 < t < \frac{T}{2} \\ \frac{A}{2} & \frac{T}{2} < t < T \end{cases}$$

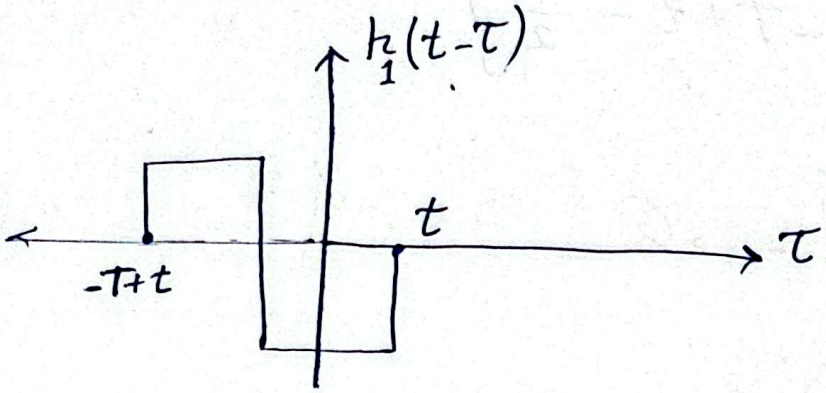
b) * MF: $y_1(t) = s_1(t) \circledast h_1(t)$

$$y_1(t) = \int_{-\infty}^{\infty} s_1(\tau)\, h_1(t-\tau)\, d\tau$$

(16)





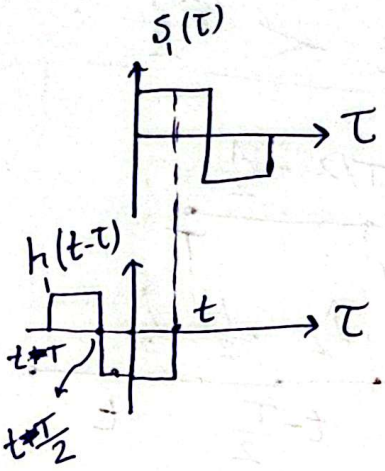


"t": 0
T/2
T
3T/2
2T

* $0 < t < \frac{T}{2}$:

$$y_1(t) = \int_0^t \left(\frac{A}{2}\right) * \left(\frac{-A}{2}\right) d\tau$$

$$\boxed{y_1(t) = \frac{-A^2}{4} t} \longrightarrow (1)$$

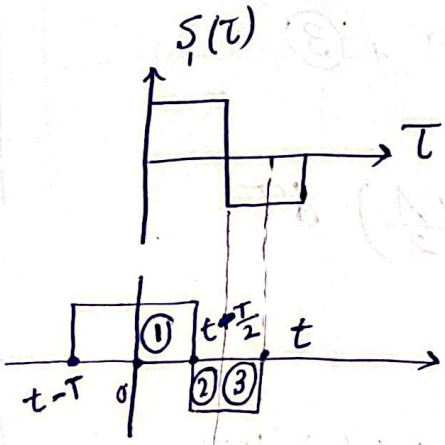


* $\frac{T}{2} < t < T$:

$$y_1(t) = \int_0^{t-\frac{T}{2}} \left(\frac{A}{2}\right)\left(\frac{A}{2}\right) d\tau \quad (1)$$

$$+ \int_{t-\frac{T}{2}}^{T/2} \left(\frac{A}{2}\right)\left(\frac{-A}{2}\right) d\tau \quad (2)$$

$$+ \int_{T/2}^{t} \left(\frac{-A}{2}\right)\left(\frac{-A}{2}\right) d\tau \quad (3)$$



(17)

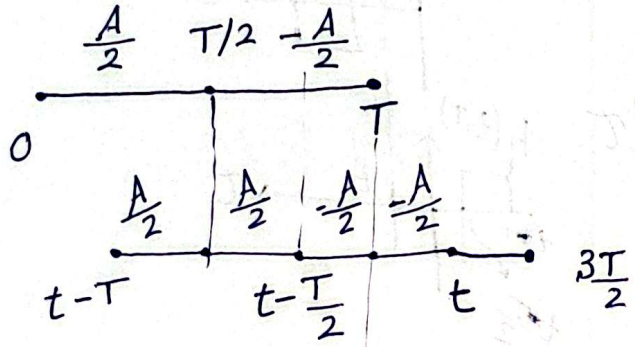
$$y_1(t) = \frac{A^2}{4}\left(t - \frac{T}{2}\right) - \frac{A^2}{4}\left[\frac{T}{2} - t + \frac{T}{2}\right] + \frac{A^2}{4}\left[t - \frac{T}{2}\right]$$

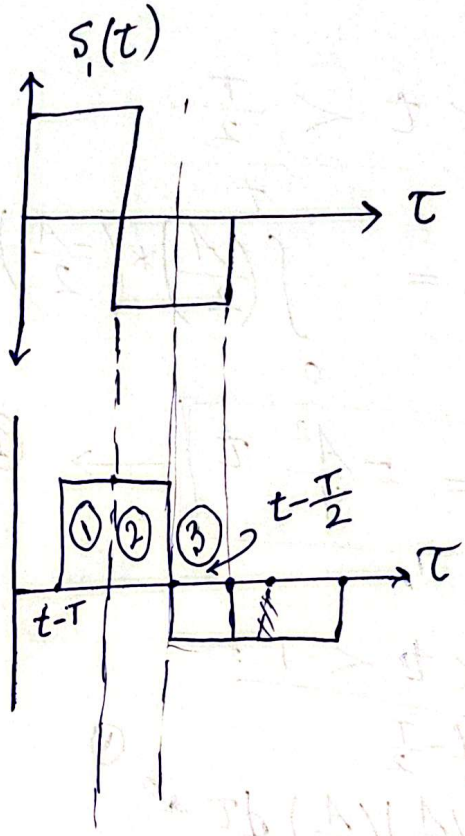
$$= \frac{A^2}{4}\left[t - \frac{T}{2}\right] + \frac{A^2}{4}\left[t - T\right] + \frac{A^2}{4}\left[t - \frac{T}{2}\right]$$

$$= \frac{A^2}{4}\left[t - \frac{T}{2} + t - T + t - \frac{T}{2}\right]$$

$$\boxed{y_1(t) = \frac{A^2}{4}(3t - 2T)} \longrightarrow (2)$$

* $T < t < \frac{3T}{2}$:





$$y_1(t) = ① + ② + ③$$

$$y_1(t) = \int_{t-T}^{T/2} \left(\frac{A}{2}\right)\left(\frac{A}{2}\right) d\tau + \int_{T/2}^{t-\frac{T}{2}} \left(\frac{-A}{2}\right)\left(\frac{A}{2}\right) d\tau + \int_{t-\frac{T}{2}}^{T} \left(\frac{-A}{2}\right)\left(\frac{-A}{2}\right) d\tau$$

(18)

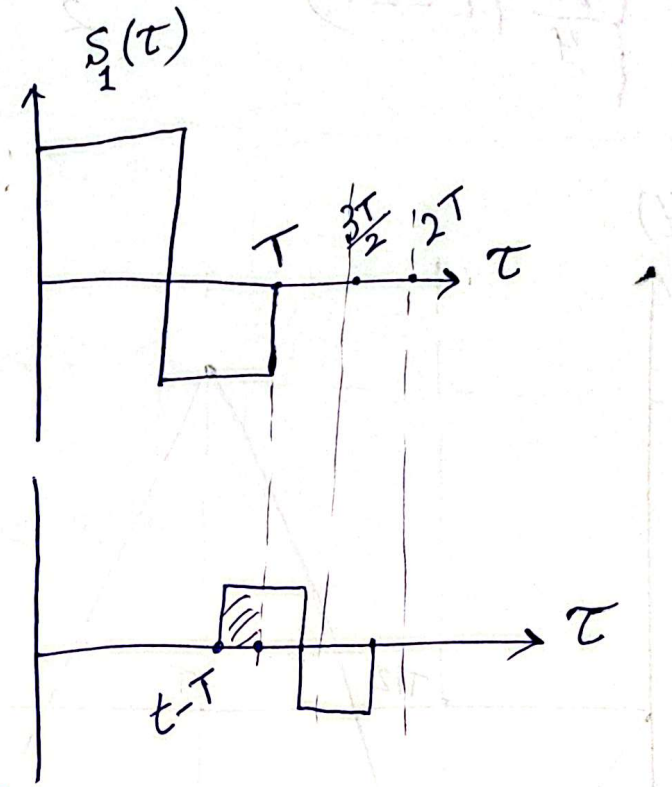
$$y_1(t) = \frac{A^2}{4}\left[\frac{T}{2} - t + T\right] - \frac{A^2}{4}\left[t - \frac{T}{2} - \frac{T}{2}\right]$$

$$+ \frac{A^2}{4}\left[T - t + \frac{T}{2}\right]$$

$$y_1(t) = \frac{A^2}{4}\left[\frac{T}{2} - t + T - t + \frac{T}{2} + \frac{T}{2} + T - t + \frac{T}{2}\right]$$

$$\boxed{y_1(t) = \frac{A^2}{4}\left[4T - 3t\right]} \rightarrow (3)$$

* $\frac{3T}{2} < t < 2T$:



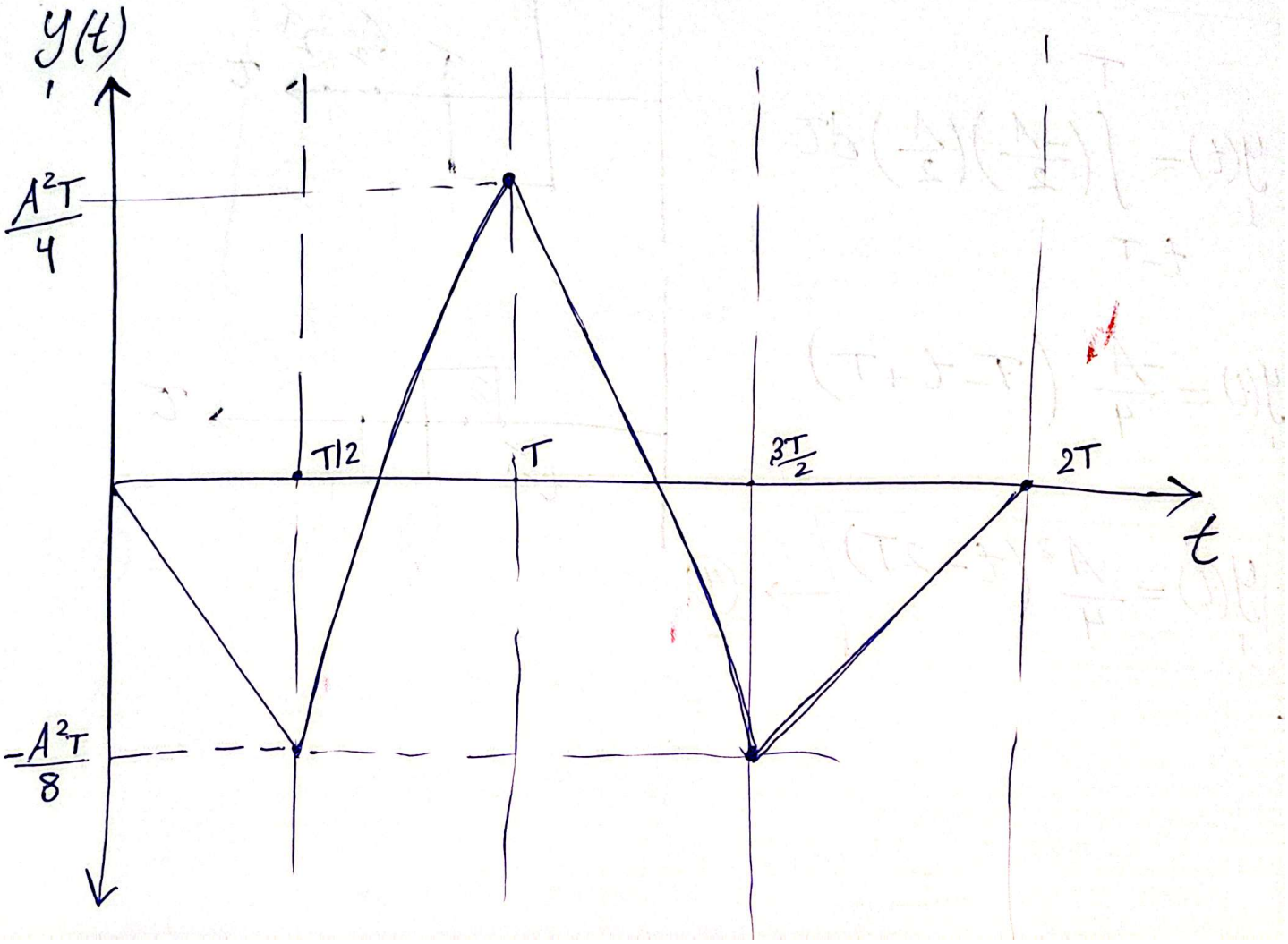
$$y_1(t) = \int_{t-T}^{T} \left(\frac{-A}{2}\right)\left(\frac{A}{2}\right) d\tau$$

$$y_1(t) = \frac{-A^2}{4}\left(T - t + T\right)$$

$$\boxed{y_1(t) = \frac{A^2}{4}(t - 2T)} \rightarrow (4)$$

From ①, ②, ③, ④:

$$y_1(t) = \begin{cases} -\frac{A^2}{4}t & 0 < t < \frac{T}{2} \\ \frac{A^2}{4}(3t - 2T) & \frac{T}{2} < t < T \\ \frac{A^2}{4}(4T - 3t) & T < t < \frac{3T}{2} \\ \frac{A^2}{4}(t - 2T) & \frac{3T}{2} < t < 2T \end{cases}$$



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$$s(t) = \begin{cases} \frac{A}{T}\cos(2\pi f_c t) & 0 < t < T \\ 0 & \text{otherwise} \end{cases}$$

Solution

a) $h(t) = s(T-t)$, $0 < t < T$

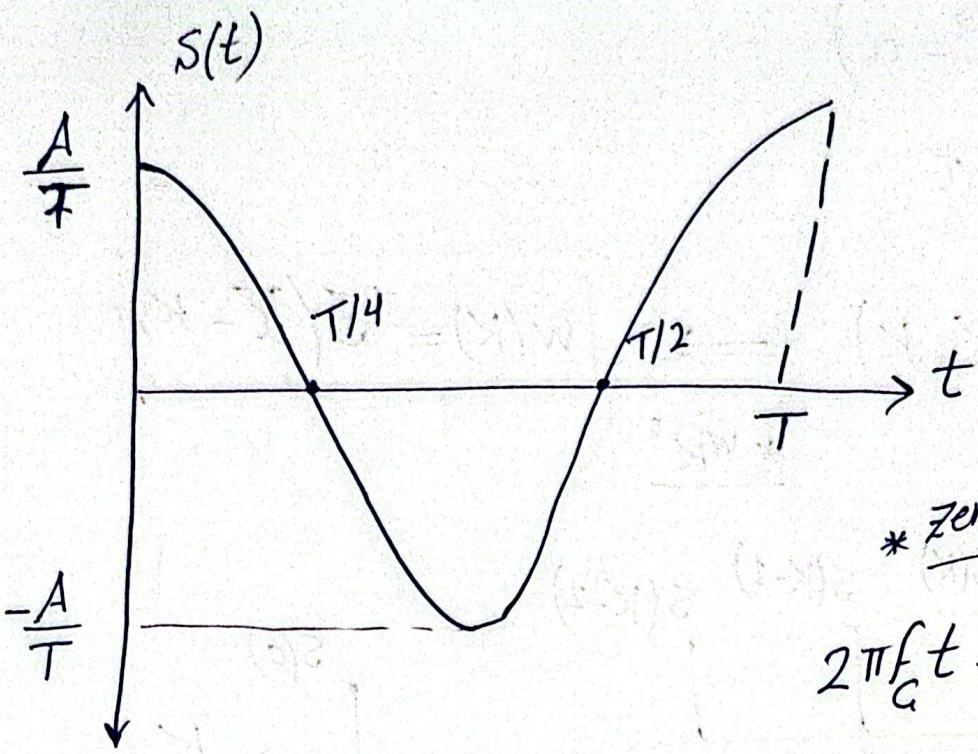
$0 < T-t < T$

$-T < -t < 0$

$0 < t < T$

$$h(t) = \begin{cases} \frac{A}{T}\cos\left(2\pi f_c (T-t)\right), & 0 < t < T \\ 0 & \text{otherwise} \end{cases}$$

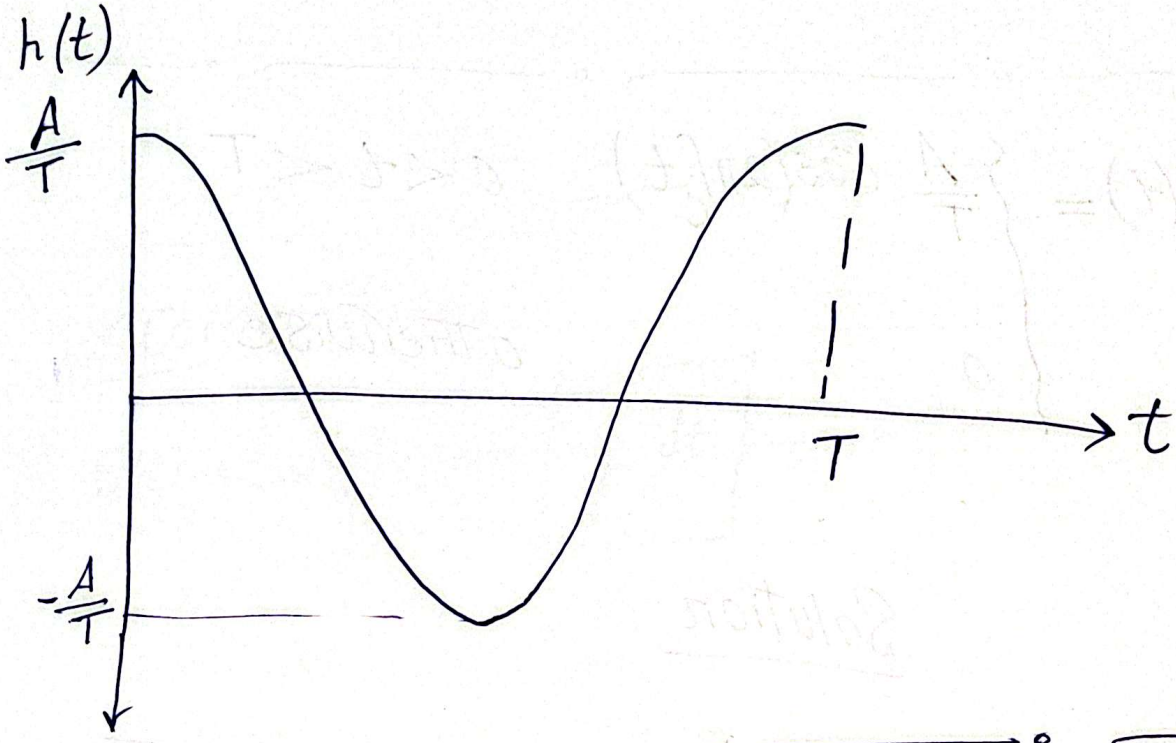
※



* Zeros of Cos:

$$2\pi f_c t = \frac{n\pi}{2}$$

$$t = \frac{n}{4f_c} = \frac{n*T}{4}$$



b) No Need for Convolution:

We know that at $t=T$, the value is the energy of the Pulse Signal

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$$E = \int_0^T \left(\frac{A}{T}\right)^2 * \cos^2(\omega_c t)\, dt \quad , \quad E = \int_0^T s^2(t)\, dt$$

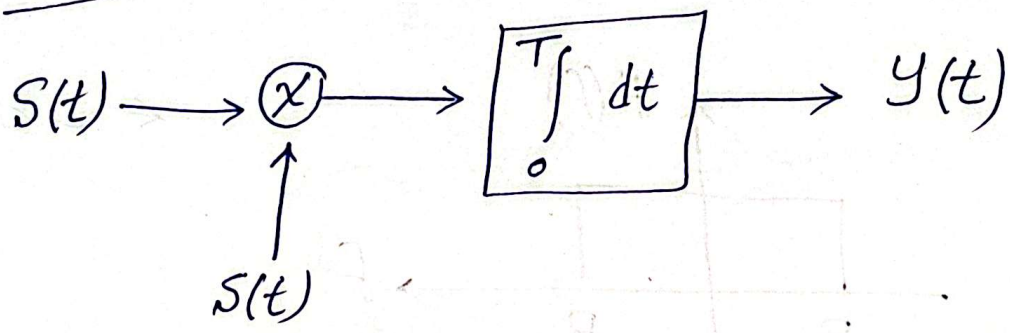
$$= \frac{A^2}{T^2} * \frac{1}{2}\left[\int_0^T 1\, dt + \int_0^T \cos(2\omega_c t)\, dt\right]$$

($\int_0^T \cos(2\omega_c t)\, dt \to 0$)

$$= \frac{A^2}{2T^2} * T = \frac{A^2}{2T}$$

$$\boxed{y(t)\Big|_{t=T} = \frac{A^2}{2T}} \ \#\#$$

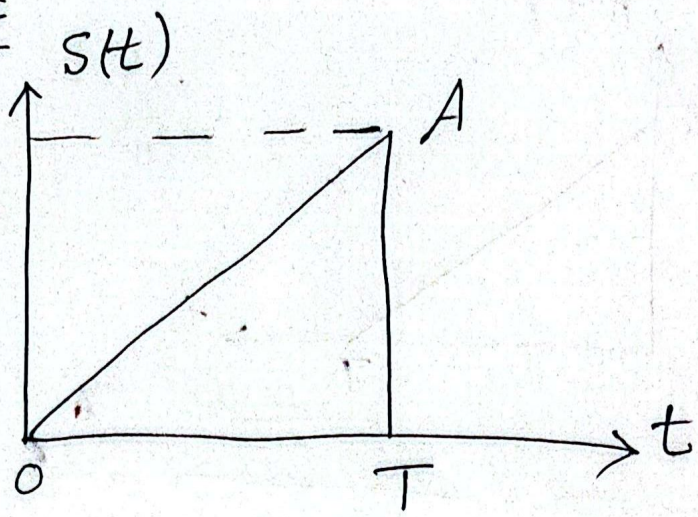
G) * Correlator:



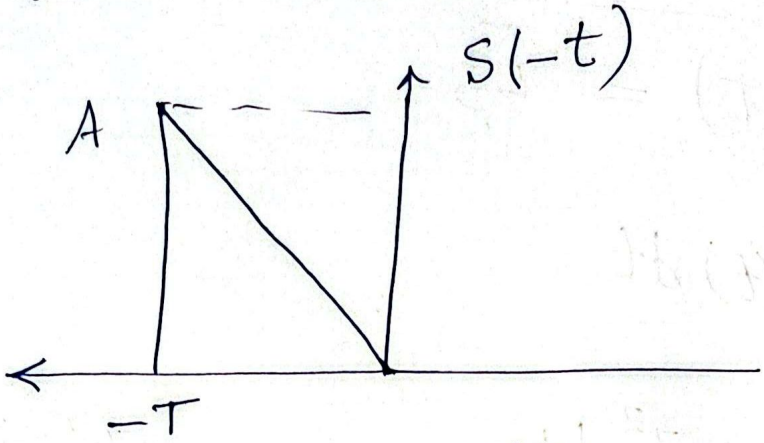
$$y(t) = \int_0^T s^2(t)\, dt = \text{Energy}$$

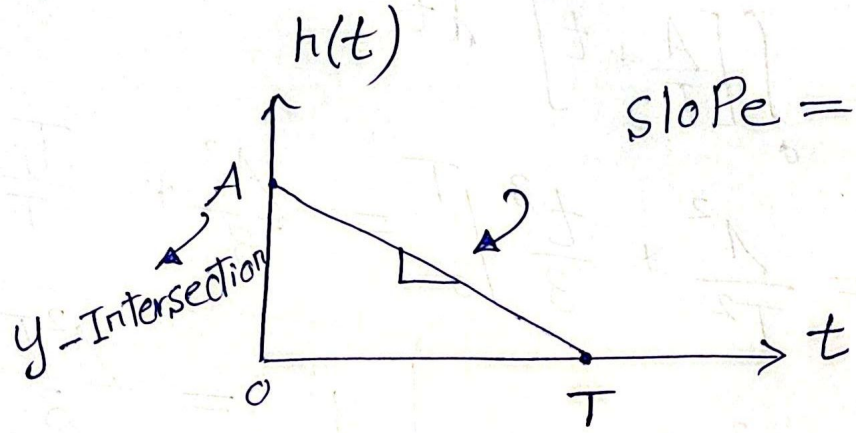
$$\boxed{y(t) = \frac{A^2}{2T}, \text{ Same Result as above.}} \ \#\#$$

Problem 3:



1) $h(t) = s(T-t)$



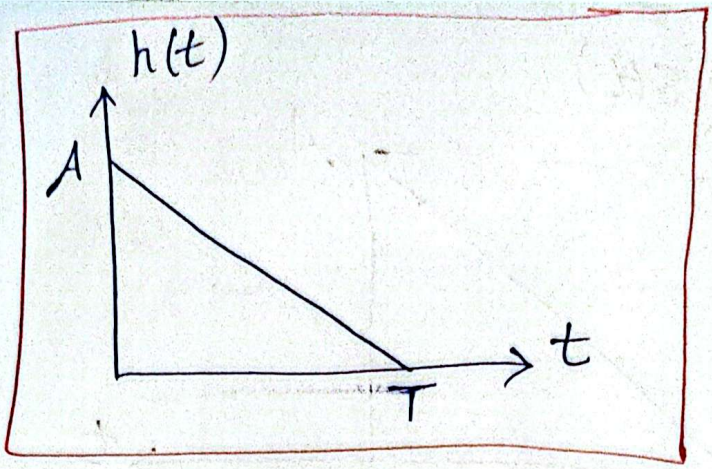


$$\text{Slope} = \frac{-A}{T}$$

Straight Line

$$h(t) = A - \frac{A}{T} * t \quad \#$$

2)



3) $y(t)_{Max}$ @ $t = T$

4) $y_{max} = y(T) = E_b$

$$E_b = \int_0^T s^2(t)\,dt$$

$$E_b = \int_0^T \left[\frac{A}{T} * t\right]^2 dt$$

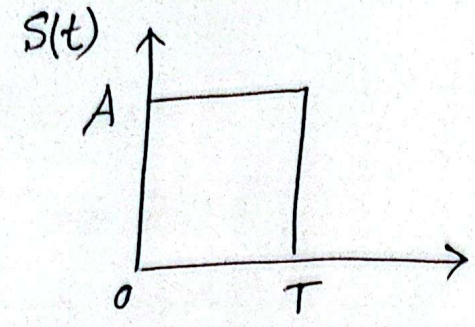
$$= \frac{A^2}{T^2} * \frac{t^3}{3}\Big|_0^T = \frac{A^2}{3} * \frac{T^3}{T^2}$$

$$E_b = \frac{A^2 * T}{3} \Longrightarrow \boxed{y_{max} = \frac{A^2 T}{3}}$$

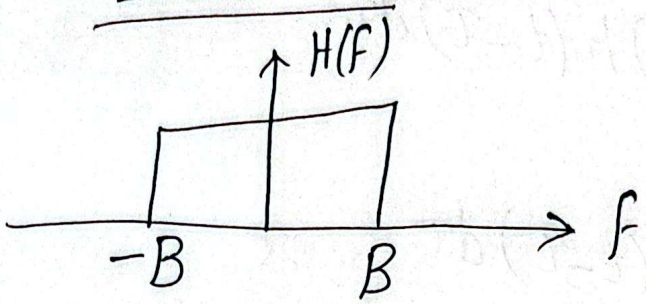
3)

*Problem 4:

$$S(t) = A \quad , \quad 0 \leq t \leq T$$

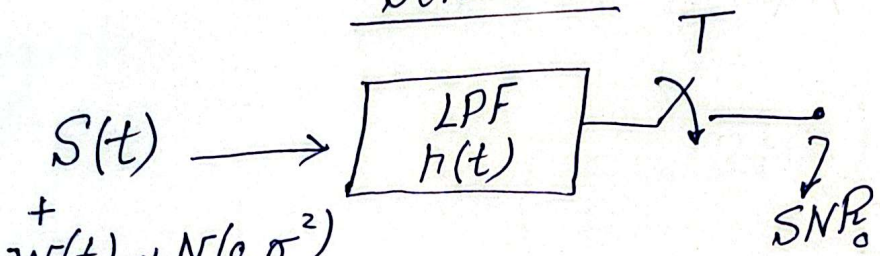


"Ideal LPF"



Solution

1)



$\sigma^2 = \frac{N_0}{2}$

$$SNR_0 = \frac{P_S}{P_N} \quad , \quad P_S = |S_0(f)|^2$$

$$\rightarrow P_N = \sigma_{n_0}^2 = \int_{-B}^{B} S_{n_0}(f)\, df$$

Noise is WSS enters LTI filter.

Then: $S_{n_0}(f) = S_{n_i}(f)\, |H(f)|^2 = \frac{N_0}{2} * 1 = \frac{N_0}{2}$

($|H(f)|^2 \sim 1$)

4)

$$P_N = \int_{-B}^{B} \frac{N_o}{2}\, df \Longrightarrow \boxed{P_N = N_o B}$$

$$\rightarrow P_S = |S_o(T)|^2$$

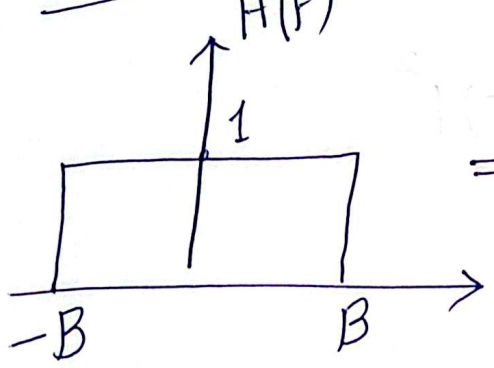
$$S_o(t) = S(t) \circledast h(t)$$

$$= \int_0^T S(\tau)\, h(t-\tau)\, d\tau$$

$$= A \int_0^T h(t-\tau)\, d\tau$$

$$S_o(T) = A \int_0^T h(T-\tau)\, d\tau$$

Get $h(t)$:



$$\Longrightarrow 2B * \text{SinC}(t * \pi * 2B)$$

$$\Downarrow$$

Normalized SinC

$$\Downarrow$$

$$2B\ \text{SinC}(2Bt)$$

$$h(t) = 2B\ \text{SinC}(2Bt)$$

$$h(T-\tau) = 2B\ \text{SinC}(2B(T-\tau))$$

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(b)

$$S_o(T) = A \int_0^T 2B \operatorname{sinc}(2B(T-\tau))\, d\tau$$

Let: $\alpha = T - \tau$, $d\alpha = -d\tau$

$\tau = 0 \Rightarrow \alpha = T$

$\tau = T \Rightarrow \alpha = 0$

$$S_o(T) = 2AB \int_0^T \operatorname{sinc}(2B\alpha)\, du$$

$$S_o(T) = 2AB \int_0^T \operatorname{sinc}(2Bu)\, du$$

Using Hint: $\operatorname{sinc}(x) \sim \cos\left(\frac{\pi}{2} x\right)$

$$\operatorname{sinc}(2Bu) = \cos\left(\frac{\pi}{2} * 2Bu\right) = \cos(\pi B u)$$

$$S_o(T) = 2AB \int_0^T \cos(\pi B u)\, du$$

$$= \frac{2AB}{\pi B} * \frac{\sin(\pi B u)}{} \Big|_0^T$$

$$\boxed{S_o(T) = \frac{2A}{\pi} \sin(\pi B T)}$$

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(6)

$$SNR = \frac{|S_0(T)|^2}{P_N}$$

$$= \frac{\frac{4A^2}{\pi^2} * \sin^2(\pi BT)}{BN_0}$$

$$SNR = \frac{4A^2}{\pi^2 BN_0} \sin^2(\pi BT)$$

→ we need to get B that Maximize SNR:

$$\frac{dSNR}{dB} = \frac{4A^2}{\pi^2 N_0}\left[\left[\frac{1}{B}\right]\left[2\sin(\pi BT) * \pi T\cos(\pi BT)\right] + \sin^2(\pi BT)\left(\frac{-1}{B^2}\right)\right]$$

$$= \frac{4A^2}{\pi^2 N_0}\left[\frac{1}{B}\sin(\pi BT)\left[2\pi T\cos(\pi BT) - \frac{1}{B}\sin(\pi BT)\right]\right]$$

$$\frac{dSNR}{dB} = 0 \implies 2\pi T\cos(\pi BT) = \frac{1}{B}\sin(\pi BT)$$

$$2\pi T B = \frac{\sin(\pi BT)}{\cos(\pi BT)}$$

$$\tan(\pi BT) = 2\pi BT$$

Let: $X = \pi BT \implies \tan(X) = 2X \implies$ Solve it Nummerically

$x \simeq 1.176 \Longrightarrow \pi BT = 1.176$

$$\boxed{B = \frac{0.374}{T}}$$

2) *LPF:

$$SNR_{LPF} = \frac{4A^2}{\pi^2 * BN_0} * \sin^2(\pi BT) \cdot \frac{T}{T}$$

$$= \frac{4A^2T}{\pi N_0} * \frac{\sin^2(\pi BT)}{\pi BT}$$

$$= \frac{4A^2}{\pi N_0} * \frac{\sin^2(x)}{x}$$

$$SNR_{LPF} = 0.9225 * \frac{A^2T}{N_0}$$

*MF: $SNR = \frac{2E_b}{N_0}$

$E_b = A^2 * T \Longrightarrow SNR_{MF} = 2 * \frac{A^2T}{N_0}$

$$\frac{SNR_{LPF}}{SNR_{MF}} = 0.46125 \xrightarrow{10\log} 3.36 \text{ dB}$$

The LPF is worse than MF with 3.36 dB

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* Problem 5:

The same Problem like "4" but with different "H(f)"

$$H(f) = \frac{1}{1 + j\frac{f}{f_0}} \quad , \quad f_0 = \frac{1}{2\pi RC}$$

$$H(f) = \frac{f_0}{f_0 + jf}$$

→ Remember: $e^{-at}u(t) \longrightarrow \frac{1}{a + j\omega}$

$$\omega = 2\pi f \Longrightarrow \frac{1}{a + j2\pi f}$$

$$H(f) = \frac{2\pi f_0}{2\pi f_0 + j2\pi f} \Longrightarrow \boxed{h(t) = 2\pi f_0 * e^{-2\pi f_0 t}u(t)}$$

→ For Noise: $P_N = \int_{-\infty}^{\infty} S_{n_0}(f)\,df$

$$S_{n_0}(f) = S_{n_i}(f)\,|H(f)|^2 \quad , \quad S_{n_i}(f) = \frac{N_0}{2}$$

$$\therefore P_N = \frac{N_0}{2} * \int_{-\infty}^{\infty} |H(f)|^2\,df$$

Parseval $\int_{\infty}^{\infty} |h(t)|^2\,dt$

S_{no}

5)

$$P_N = \frac{N_0}{2} * \int_0^{\infty} (2\pi f_0)^2 * e^{-4\pi f_0 t} \, dt$$

$$= 2\pi^2 f_0^2 N_0 * \left[\frac{e^{-4\pi f_0 t}}{-4\pi f_0} \right]_0^{\infty}$$

$$= 2\pi^2 f_0^2 N_0 * \frac{1}{4\pi f_0}$$

$$\boxed{P_N = \frac{\pi f_0 N_0}{2}}$$

$$\rightarrow P_S = |S_0(t)|^2$$

$$S_0(t) = S(t) \circledast h(t)$$

$$= \int_{-\infty}^{\infty} S(\tau)\, h(t-\tau)\, d\tau$$

$$S_0(T) = A \int_0^T 2\pi f_0 \, e^{-j2\pi f_0 (t-\tau)} \, d\tau$$

$$S_0(t) = 2\pi f_0 A \int_0^T e^{-j2\pi f_0 x} \, dx$$

(10)

$$S_0(T) = 2\pi f_0 A * \frac{e^{-j2\pi f_0 x}}{-2\pi f_0}\Big|_0^T$$

$$\boxed{S_0(T) = A\left[1 - e^{-2\pi f_0 T}\right]}$$

$$SNR = \frac{A^2\left[1 - e^{-2\pi f_0 T}\right]^2}{\frac{\pi f_0 N_0}{2}}$$

$$= \frac{2A^2}{N_0 \pi f_0}\left[1 - e^{-2\pi f_0 T}\right]^2$$

$$\frac{dSNR}{df_0} = \frac{2A^2}{N_0\pi}\left[\left[\frac{1}{f_0}\right]\left[2(1 - e^{-2\pi f_0 T}) * e^{-2\pi f_0 T} * 2\pi T\right] + \left[1 - e^{-2\pi f_0 T}\right]^2 * \frac{-1}{f_0^2}\right]$$

$$= \frac{2A^2}{N_0\pi} * \left[\frac{1}{f_0}\left(1 - e^{-2\pi f_0 T}\right)\left[4\pi T e^{-2\pi f_0 T} - \frac{1}{f_0}\left(1 - e^{-2\pi f_0 T}\right)\right]\right]$$

$$\frac{dSNR}{df_0} = 0 \Longrightarrow \frac{1}{f_0}\left(1 - e^{-2\pi f_0 T}\right) = 4\pi T e^{-2\pi f_0 T}$$

$$f_0 = \frac{1 - e^{-2\pi f_0 T}}{4\pi T e^{-2\pi f_0 T}}$$

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Let: $x = 2\pi f_0 T$

$$f_0 = \frac{1 - e^{-x}}{2(2\pi T)\, e^{-x}}$$

$$2x e^{-x} = 1 - e^{-x}$$

$$x = 1.25643$$

$$2\pi f_0 T = 1.25643 \Longrightarrow \boxed{f_0 \simeq \frac{0.2}{T}} \ \#\#$$

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2) * LPF:

$$SNR_{LPF} = \frac{2A^2}{N_0 \pi f_0} * \left(1 - e^{-2\pi f_0 T}\right)^2 * \frac{2T}{2T}$$

$$= \frac{4A^2 * T}{N_0 2\pi f_0 T} * \left(1 - e^{-2\pi f_0 T}\right)^2$$

$$SNR_{LPF} = \frac{4A^2 T}{x} * \left(1 - e^{-x}\right)^2 = 1.629 \frac{A^2 T}{N_0}$$

* MF: $SNR = \frac{Eb}{2N_0}$

$$SNR_{MF} = 2 * \frac{A^2 * T}{N_0}$$

$$\frac{SNR_{IPF}}{SNR_{MF}} = 0.8145$$

There is Loss in SNR with 0.89 $\sim$ 1 dB.